Progress Report — Week 2

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Report return date: 15 May 2026

Reporting period

8 May 2026 – 14 May 2026

Hours worked

Date	Day	Hours	Activity summary
8 May 2026	Fri	8	Architecture design, module dependency mapping
11 May 2026	Mon	8	Baseline sprint simulation loop implementation
12 May 2026	Tue	8	Team member dataclass and team generation
13 May 2026	Wed	8	Backlog/task generation and sprint task selection
14 May 2026	Thu	8	Code review with supervisor, refactor module interfaces
Total		**40**	

Work completed

- Designed four-layer architecture: UI, orchestration, simulation engine, configuration.
- Implemented the main per-sprint loop in `simulation.py` (task selection → allocation → completion/defects → CI update).
- Created `TeamMember` dataclass and team generation logic in `team.py`.
- Implemented backlog generation and sprint task selection in `tasks.py`.
- Ran first end-to-end smoke test of a single sprint with fixed random seed.

Connection to learning objectives

- Applied software architecture and object-oriented design from university courses.

- Worked independently on implementation while aligning design decisions with supervisor expectations.
- Used systematic debugging to validate module interactions.

Challenges and how they were handled

Early coupling between task selection and metrics made testing difficult. I separated interfaces and added clearer function boundaries so each module could be tested in isolation.

Plan for next week (15–21 May 2026)

- Implement CI subconstruct formulas in `metrics.py`.
- Build AI support modules for allocation and shared cognition.
- Connect metrics output to simulation results and basic Streamlit display.

Student signature: Arslan Sultan — Date: 15 May 2026
Supervisor acknowledgment: Juan Garbajosa